

Waleed Sherwani

wmksherwani@mun.ca ❖ +1 (709) 327-5285 ❖ St. John's, NL, CA | [GitHub](#) | [LinkedIn](#)

EDUCATION

Memorial University of Newfoundland (MUN)

2022 – 2026

Bachelor of Engineering

St. John's, NL, Canada

- Computer Engineering | Class of 2026 | GPA: 3.80
 - Faculty of Engineering and Applied Sciences: Dean's list (2022 – 2023)

WORK EXPERIENCE

Data Platform Operations Developer

May. 2026 – Present

Nasdaq

St. John's, NL, Canada

- Support production data platform operations across Data Lake, Data Warehouse, and ETL/ELT environments, monitoring pipeline health, investigating failed jobs, and resolving missing or delayed customer data workflows.
- Develop and maintain cloud-native data platform infrastructure using Terraform, AWS Lambda, S3, Athena, Step Functions, Spark, Scala, Java, Python, and SQL to support reliable customer data onboarding and movement.
- Debug production incidents involving Spark job failures, AWS service configurations, access permissions, deployment errors, & performance bottlenecks, collaborating with engineering teams to restore pipeline reliability.
- Contribute to platform reliability initiatives by improving monitoring, operational visibility, and automation for customer data pipelines, while onboarding into formal on-call responsibilities for production support.

AI & Automation Developer Intern

Sep. 2025 – April. 2026

Nasdaq

St. John's, NL, Canada

- Designed the end-to-end architecture for an AI “Mixture-of-Experts” service that consumes code changes and surrounding organizational context, orchestrating multiple specialized LLM agents (e.g., performance, risk, testing, code quality) to generate feedback directly within GitHub workflows.
- Developed cloud-native data and analytics pipelines that aggregate engineering and knowledge-base telemetry from internal Q&A, code hosting, documentation, and CRM platforms into a centralized Data Lake on AWS, enabling SPACE and DORA-style productivity dashboards in Grafana.
- Implemented serverless collectors and ETL workflows using AWS services (e.g., Lambda, API Gateway, Glue, S3, Athena) to reliably ingest, transform, and surface engineering metrics for self-serve analytics and reporting.
- Built a retrieval and context-routing layer that blends structured metrics, knowledge-base content, and vectorized embeddings to ground agent reasoning and improve the relevance of AI-generated recommendations.

Datalake Developer Intern

May. 2024 – Aug. 2024 & Jan. 2025 – Aug. 2025

Nasdaq

St. John's, NL, Canada

- Contributed to the development of a new data lake pipeline, utilizing Java, Scala, Spark, and Serverless Computing on AWS, with a strong focus on data storage and pre-processing.
- Implemented database splitting logic in Scala code and maintained setups for new data types, ensuring data quality through comprehensive data validations using Spark and Jupyter Notebooks.
- Collaborated with a team of data analysts to develop and implement data analytics solutions.
- Made significant contributions to the ongoing development of the Step Function in the Data Pipeline, integrating it with other AWS services to enhance pipeline efficiency.

Data Analytics and Project Control Intern

Sep. 2023 – Dec. 2023

SNC-Lavalin - Dragados - Pennecon General Partnership (SDP-GP)

Argentina, NL, Canada

- Programmed specialized data analytics workflows in Python and C++, using libraries Pandas, NumPy, and Matplotlib for data manipulation, validation, and visualization, enhancing data processing and analysis.
- Implemented web parsing techniques for efficient database data collection using Selenium & Requests-HTML
- Automated File-Sorting and Document Generation using Image Recognition with Tesseract and TensorFlow, based on a keyword model developed on a trained set of documents.

Robotics Research Intern

Memorial University of Newfoundland

Jan. 2023 – May. 2023

St. John's, NL, Canada

- Integrated DekaBot & Adafruit Color Sensors into ZumoBots and developed custom C++ software to operate them.
- Developed ZumoBots' implementation in Swarm Robotics by comparing evaluated improvements from the DekaBot.
- Programmed contextual Line Detection Software and Alignment Algorithms with PID Control loop mechanisms.
- Implemented I²C protocol for communication between ZumoBots' microcontroller and various sensors and modules, improving the functionality and efficiency of the robots.
- Tested and debugged sensors and actuators in design implementations.

ADDITIONAL EXPERIENCE

Valiant Aerotech – Software Team Lead

Student Design Hub – Memorial University of Newfoundland

Sep. 2024 – Present

St. John's, NL, Canada

- Lead a software team of 5 engineers in the design, development, and deployment of autonomous flight control systems for UAV platforms.
- Represented Valiant Aerotech at the 16th AEAC Annual Competition in Alberta, showcasing autonomous flight prototyping and sensor-fusion techniques for firefighting applications.

ISWEP Scientific Computing – Research Assistant

Department of Chemistry – Memorial University of Newfoundland

Sep. 2024 – Feb 2025

St. John's, NL, Canada

- Designed specialized implementation for π bonded molecular generation on the CReM framework.
- Developed Jupyter Notebooks with data analytics implementation for researchers.

Data Quality Specialist

Medical Document Team – SiftMed

May. 2024 – Aug. 2024

St. John's, NL, Canada

- Performed comprehensive QA for system output, ensuring accurate document tags, groupings, and metadata before final client delivery.
- Automated document processing tasks, streamlining workflows with keyboard commands and resolving duplicate documents efficiently, using python

Data Quality Assurance Analyst

Data and Image Analysis Group - Memorial University of Newfoundland

May. 2023 – Sep. 2023

St. John's, NL, Canada

- Analyzed medical images, ensuring data accuracy, and collaborating with researchers from both the DIAG and FAISAL Labs to improve patient care through automated image-derived metrics.

ISWEP Research Assistant

Graduate Office – Memorial University of Newfoundland

Jun. 2022 – Sep. 2022

St. John's, NL, Canada

- Programmed a Python script that parses data from an Excel workbook and performs a selective Internet Search.
- Designed a survey analysis format using Excel, with automatic, predefined data visualizations such as linear models, predicted changes, and bell curves.

PROJECTS

- More on [Github](#) :)

SKILLS

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|---------|--------|------------|----|------------|---------|-----------|----------|-----|
| ▪ C++ | Python | Java | C# | JavaScript | Django | Flask | MongoDB | SQL |
| ▪ NumPy | Pandas | Matplotlib | | TensorFlow | PyTorch | Tesseract | Selenium | BS4 |
| ▪ KiCAD | PSpice | MATLAB | | AutoCAD | Blender | MS Office | Assembly | R |

CERTIFICATIONS

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|---|-----------|
| ▪ <u>Google Advanced Data Analytics</u> | Sep. 2023 |
| ▪ <u>Google Data Analytics</u> | Jan. 2023 |
| ▪ <u>Onshape Fundamentals: CAD Learning Pathway Completion</u> | Jan. 2023 |
| ▪ <u>Tech Stewardship Practice Program (TSPP)</u> | Dec. 2022 |
| ▪ <u>Workplace Hazardous Materials Information System (WHMIS)</u> | Aug. 2022 |